Accessible Observable Contrast: Symmetry-Resolved Witnesses, Local No-Go Certificates, and Topological Flux

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***Abstract*—A discriminative observable can be physically inaccessible because it depends on an arbitrary reference frame, violates a symmetry, or exceeds the locality of an instrument. We formulate maximum observable contrast inside a declared finite-dimensional observable algebra. If $\mathcal{E}_{\mathcal{A}}$ is the trace-preserving conditional expectation onto a unital $*$-subalgebra $\mathcal{A}$, the largest expectation gap over effects in $\mathcal{A}$ is $\mathrm{Tr}[\mathcal{E}_{\mathcal{A}}(\rho_1 - \rho_0)_+]$. Group twirling, fixed-region readout, and representation-sector resolution follow as exact cases; class states are additive and mergeable. We retain controlled validations in translation-invariant signals, an Ising reduced state, a Hückel difference density, and robot contact, including ties and negative controls. We then perform an exact 18-qubit toric-code benchmark. All 1431 Pauli observables below code distance and all 171 one- or two-link reduced-state comparisons are strictly blind to the selected topological-flux label. Among 22032 weight-three Pauli strings, exactly three homologous Wilson loops have nonzero class gap. A label-preserving logical twirl changes one-representative-per-class inference from 0.75 to 1, and the learned code-space effect equals the Wilson-sector projector; a label-flipping twirl returns chance. Helstrom discrimination, a supplied Wilson threshold, and the learned effect all tie at one. Thus the result establishes an observable-access hierarchy, a symmetry-prior benefit, and a no-information certificate—not a universal algorithmic or quantum advantage.**

***Index Terms*—accessible observables, additive state estimation, lattice gauge theory, quantum state discrimination, symmetry resolution, topological order, Wilson loops**

I. Introduction

An unconstrained detector answers the wrong question when an observer lacks an absolute phase, only a regional instrument is available, or gauge-variant readouts are unphysical. Data augmentation may approximate an invariance, but it does not identify the best bounded effect that is actually available, nor certify that an entire measurement class contains no information.

Restricted quantum measurements and invariant hypothesis testing are established subjects [1], [2]. Harmonic analysis yields the associated representation blocks [3], [4], while symmetry-resolved entanglement decomposes reduced states by global charge sectors [5]. We combine these foundations with additive positive-state summaries under the name symmetry-/subalgebra-accessible observable contrast (SAOC). The synthesis is deliberately broader than a group: a fixed regional algebra and block readout are algebraic examples, while bandwidth, bounded-weight, and other feature dictionaries define restricted convex problems that need not be algebras.

Working paper targeting venue fit with IEEE Transactions on Quantum Engineering. All numerical claims are backed by fixed scripts, raw records, tests, and manifests in the accompanying public repository. The chemistry and robotics results are controlled models, and the gauge result is an exact finite fixed-point benchmark, not a continuum or holographic claim.

The new question in this run is whether the same formalism can expose topological information that every sufficiently local observer provably misses. The toric code supplies a sharp test: orthogonal ground sectors are locally indistinguishable, while a noncontractible Wilson loop resolves their global flux [6], [7]. Numerical searches for ribbon operators [8] and machine learning of topological order [9] already exist. We therefore do not present Wilson loops or local topological order as discoveries.

This paper contributes:

1) an accessible-contrast theorem and compact-group corollary, together with additive and mergeable class-state estimation;
2) sector diagnostics and a fixed-region no-information certificate;
3) reproducible Run 3 tests in invariant signals, many-body response, difference-density chemistry, and robot contact with explicit strong baselines and limitations;
4) an exhaustive $L = 3$ toric-code test that moves from exact local blindness to the first discriminative noncontractible support and checks correct and wrong symmetry priors;
5) a quantum implementability and scaling audit that distinguishes a compilable structured witness from a general exponentially large projector.

II. Accessible-Observable Theory

A. Unrestricted and algebra-restricted contrast

Let $\rho_0, \rho_1 \succeq 0$ have unit trace and $\Delta = \rho_1 - \rho_0$. Over all effects $0 \preceq E \preceq I$,

$$\max_E \mathrm{Tr}(E\Delta) = \mathrm{Tr}(\Delta_+) = \tfrac{1}{2}\|\Delta\|_1, \tag{1}$$

with $E^\star = \mathbf{1}_{\Delta>0}$. This is the Helstrom–Jordan result [10], not a new quantum theorem.

Let $\mathcal{A} \subseteq M_d(\mathbb{C})$ be a unital $*$-subalgebra and let $\mathcal{E}_{\mathcal{A}}$ be its trace-preserving conditional expectation.

Theorem 1 (accessible contrast).

$$\begin{aligned} D_{\mathcal{A}}(\rho_1, \rho_0) &:= \sup_{\substack{E\in\mathcal{A} \\ 0\preceq E\preceq I}} \mathrm{Tr}(E\Delta) \\ &= \mathrm{Tr}[\mathcal{E}_{\mathcal{A}}(\Delta)_+] = \tfrac{1}{2}\|\mathcal{E}_{\mathcal{A}}(\Delta)\|_1. \end{aligned} \tag{2}$$

An optimizer is the positive support projector of $\mathcal{E}_{\mathcal{A}}(\Delta)$, formed inside $\mathcal{A}$.

Proof: For $E \in \mathcal{A}$, Hilbert–Schmidt orthogonal projection gives $\mathrm{Tr}(E\Delta) = \mathrm{Tr}[E\mathcal{E}_{\mathcal{A}}(\Delta)]$. Conditional expectation preserves Hermiticity and maps into $\mathcal{A}$. The Jordan bound gives the upper bound, and functional calculus places the positive support projector in $\mathcal{A}$, attaining it. ■

The restricted and unrestricted optima coincide exactly when an effect in $\mathcal{A}$ acts as one on the positive support of Δ and as zero on its negative support; its action on $\ker\Delta$ is arbitrary.

Equation (2) does not apply unchanged to an arbitrary feature dictionary that is not closed under products and adjoints; the corresponding restricted convex problem must then be solved directly.

B. Group symmetry and sectors

For a compact group with representation U_g, the commutant $\mathcal{A} = \{U_g\}'$ contains invariant observables, and

$$\mathcal{T}_G(X) = \int_G U_g X U_g^\dagger \, dg \tag{3}$$

is the conditional expectation. Hence

$$\max_{\substack{0\preceq E\preceq I \\ [E,U_g]=0}} \operatorname{Tr}(E\Delta) = \operatorname{Tr}[\mathcal{T}_G(\Delta)_+]. \tag{4}$$

If

$$\mathcal{H} = \bigoplus_\lambda V_\lambda \otimes M_\lambda, \tag{5}$$

then Schur's lemma yields

$$\mathcal{T}_G(\Delta) = \bigoplus_\lambda \frac{I_{V_\lambda}}{\dim V_\lambda} \otimes \Delta_\lambda. \tag{6}$$

Here P_λ projects onto $V_\lambda \otimes M_\lambda$ and $\Delta_\lambda = \operatorname{Tr}_{V_\lambda}(P_\lambda \Delta P_\lambda)$. Trace norm and positive gap add over the representation sectors.

For cyclic translations, the DFT diagonalizes every invariant operator:

$$\mathcal{T}_{C_d}(|x\rangle\langle x|) = F^\dagger \operatorname{diag}(|Fx|^2) F. \tag{7}$$

The specialized vector implementation requires $O(d \log d)$ time and $O(d)$ state per signal. The repository's general convenience path still materializes dense matrices, so this theoretical special-case complexity is not reported as a measured implementation advantage.

C. Additive and mergeable estimation

For encoded positive samples $\sigma(x) \succeq 0$, class c maintains

$$\begin{aligned} S_{c,t} &= S_{c,t-1} + w_t \sigma(x_t), \\ W_{c,t} &= W_{c,t-1} + w_t, \qquad \widehat{\rho}_{c,t} = S_{c,t}/W_{c,t}. \end{aligned} \tag{8}$$

Pairs (S, W) merge by componentwise addition, so distributed sites need not retain raw samples. A decay factor can track drift; an exact sliding window additionally stores the samples to be deleted. General dense states cost $O(d^2)$ memory, whereas diagonal, low-rank, stabilizer, or tensor-network structure may reduce that cost.

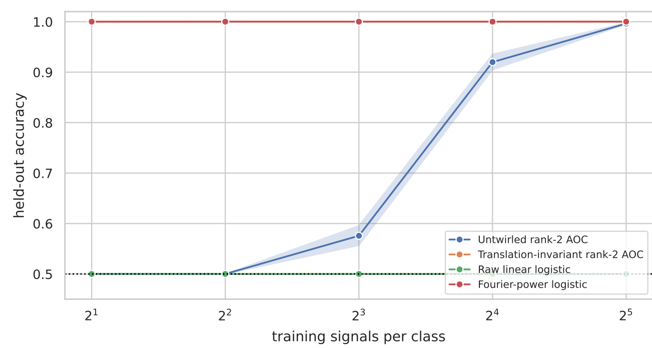

Fig. 1. Run 3 translation nuisance. Invariant contrast and the correctly engineered Fourier baseline solve the task with one sign-pair per class; the baseline prevents a universal-algorithm claim.

D. Fixed regions, local blindness, and gauge structure

For a fixed link region R, the full operator algebra on R is a genuine $*$-subalgebra after extension by the identity outside R. Its accessible contrast is the trace distance of the two reduced states on R. If P is a quantum-code projector and R is correctable, the Knill–Laflamme/local-topological-order condition is

$$P O_R P = c_{O_R} P \quad \text{for every operator supported on } R. \tag{9}$$

It follows that all code states have the same ρ_R and $D_R = 0$ [7], [11]. This is a no-information certificate, not a classifier-specific failure.

Gauge theories require an additional convention: the physical Hilbert space does not generally factor into complementary spatial tensor products, and regional gauge-invariant algebras can have centers [12], [13]. Our ordinary link-qubit partial trace is the extended-link (electric-center) prescription. A global-symmetry charge sector, a regional boundary-flux sector, and a torus topological holonomy are related block structures but are not identical.

III. Controlled Run 3 Evidence

Every result has raw CSV/JSON, a deterministic command, package versions, hashes, and tests. These controlled examples test distinct consequences of the same operator construction rather than claim universal superiority.

A. Translation nuisance and quantum response

Length-128 signals at DFT bins 5 and 17 have random cyclic shift, random phase, Gaussian noise 0.8, and exact sign partners. Over 12 seeds and 600 test signals per class, two training signals per class give accuracy 0.999861 for invariant contrast, 1.0 for Fourier-power logistic, and 0.5 for both raw linear logistic and untwirled contrast. The correctly specified Fourier baseline wins by 1.39×10^{-4}; the result isolates the symmetry prior rather than a unique SAOC advantage.

A ten-qubit periodic transverse-field Ising chain is exactly diagonalized, five qubits are traced out, and adjacent reduced states are compared on a $\delta g = 0.025$ grid. The response peaks at $g/J = 0.9625$, near the thermodynamic critical

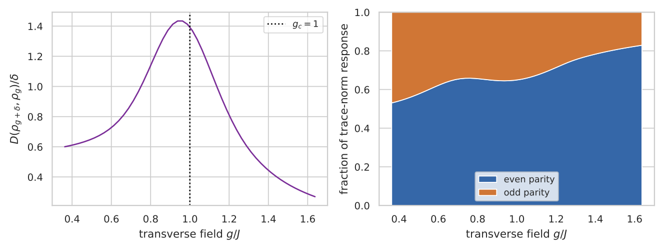

Fig. 2. Run 3 Ising reduced-state response and parity-sector decomposition. The dotted line is the thermodynamic critical field.

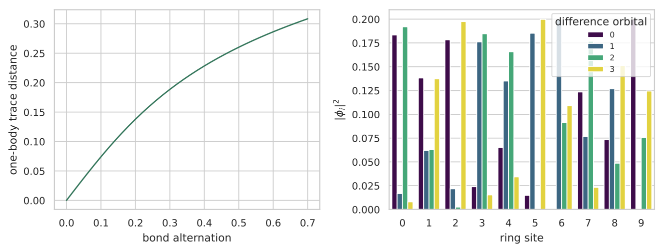

Fig. 3. Run 3 controlled Hückel difference density and leading natural difference orbitals.

value one, and the maximum parity commutator residual is 3.54×10^{-9}. This is a finite-size diagnostic, not a new critical exponent [14]–[16].

B. Difference-density chemistry and robot contact

A half-filled ten-site Hückel ring is changed from uniform to alternating hopping. At alternation 0.7, its one-body trace distance is 0.30845, and the positive and negative spectral parts define attachment and detachment contributions whose weights balance to 5.55×10^{-17}. Difference-density natural orbitals are established chemical analysis tools [17], [18]; this is a controlled conservation test, not quantitative molecular prediction.

A zero-mean six-axis force/torque residual receives a planted rank-one contact covariance along a wrench screw. At 64-sample windows, the learned and oracle scores both have ROC AUC 1.0, mean logistic is 0.5 by construction, and learned/oracle screw overlap is 0.99925. Small-window results trail the oracle, and the experiment has neither hardware data nor an equal-budget sample-efficiency claim.

IV. Run 4: Topological-Flux Experiment

A. Exact model and exhaustive design

On a periodic $L \times L$ square lattice, place one qubit on each of $2L^2$ oriented links and define

$$A_s = \prod_{e \ni s} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e,$$
$$H = -\sum_s A_s - \sum_p B_p. \tag{10}$$

For $L = 3$, the 18-qubit code is $[[18, 2, 3]]$. We construct all four exact ground sectors from the star group rather than diagonalize a degenerate Hamiltonian. Every state has 256

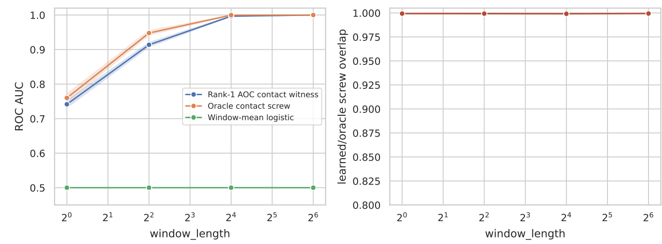

Fig. 4. Run 3 controlled contact result. The rank-one quadratic witness recovers the planted wrench direction; this is not deployment evidence.

TABLE I
Exhaustive Accessibility Hierarchy for the Selected Flux Label.

Candidate class	Count	Nonzero contrast
Weight-1 Pauli strings	54	0
Weight-2 Pauli strings	1377	0
Weight-3 Pauli strings	22032	3
One-link RDM supports	18	0
Two-link RDM supports	153	0
Three-link RDM supports	816	3

equal-magnitude computational-basis amplitudes and satisfies $A_s = B_p = +1$. We fix $\bar{Z}_y = +1$ and classify the $\bar{Z}_x = \pm 1$ sectors.

We exhaust all Pauli strings of weights one through three and all link subsets of sizes one through three. No $2^{18} \times 2^{18}$ density matrix is formed: dense state vectors with 256-point sparse support and selected reduced matrices are used. The state-vector dimension is 262144, so this run is a finite exact benchmark, not a scaling demonstration.

Table I verifies exact blindness below distance. The three nonzero weight-three strings are the homologous horizontal ZZZ Wilson loops, with expectation gap two. Twelve weight-three Pauli strings commute with all stabilizers, but only those three carry the selected label. For any one loop R,

$$\rho_{\pm}^R = \frac{P_{\pm}}{2^{L-1}}, \qquad P_{\pm} = \frac{I \pm W}{2}, \qquad \tfrac{1}{2}\|\rho_+^R - \rho_-^R\|_1 = 1. \tag{11}$$

B. Logical nuisance symmetry and controls

In the logical basis $|a, b\rangle$, a is the $\bar{Z}_x$ label and b is an unknown $\bar{Z}_y$ nuisance. Training uses one exact representative state description $|0, 0\rangle$ and $|1, 0\rangle$ per class; this is not tomography from one unknown quantum copy. Twirling over the label-preserving group $G_{\text{nuis}} = \{I, \bar{X}_y\}$ gives

$$\rho_+ = \tfrac{1}{2}\,\text{diag}(1,1,0,0), \qquad \rho_- = \tfrac{1}{2}\,\text{diag}(0,0,1,1), \tag{12}$$

and, on the code space,

$$\Delta = \tfrac{1}{2}\bar{Z}_x, \qquad \text{sign}(\Delta) = \bar{Z}_x, \qquad E^\star = \frac{I + \bar{Z}_x}{2}. \tag{13}$$

The effect outside the supported code space is not identified and is not compared naively with a full-Hilbert-space Wilson operator.

The 0.75 untwirled value uses the canonical strict-positive support and sets the unseen kernel to zero. A kernel completion

TABLE II
EQUAL-PRIOR FLUX-SECTOR INFERENCE UNDER MATCHED TEST NUISANCE.

Method/access model	Success
Any fixed ≤ 2-link observer	0.50
Untwirled one-representative SAOC	0.75
No nuisance twirl (stabilized representatives)	0.75
Correct nuisance twirl + SAOC	1.00
Wrong label-flipping twirl + SAOC	0.50
Supplied Wilson-parity threshold	1.00
Full Helstrom oracle	1.00

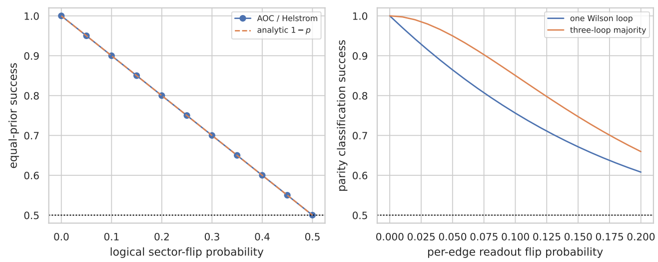

Fig. 5. Run 4 analytic calibration. Left: SAOC and Helstrom coincide under logical-sector mixing. Right: redundant homologous-loop majority improves readout robustness at additional measurement cost.

supplied with the missing logical prior can attain one, so 0.75 is a reproducible ablation rather than an information-theoretic upper bound. The "no nuisance twirl" row is an identity action in logical space because both representatives already satisfy all stabilizers. The deliberately wrong group $\{I, \bar{X}_x\}$ flips the label, makes the two twirled states equal, and returns chance. Correct twirling therefore removes a nuisance; it does not manufacture information. It also cannot increase unrestricted trace distance.

C. Noise calibration

Logical-sector mixing with probability p gives

$$\widetilde{\rho}_\pm(p) = (1-p)\rho_\pm + p\rho_\mp, \quad D(p) = |1-2p|, \quad P_{\text{succ}} = 1-p \tag{14}$$

for $0 \leq p \leq 1/2$. All 11 numerical points match Eq. (14) exactly. With independent per-edge readout flip probability r, a length-three parity is correct with

$$q(r) = \frac{1 + (1-2r)^3}{2}. \tag{15}$$

Majority vote over three disjoint homologous loops has success $3q^2 - 2q^3$. At $r = 0.05$, the values are 0.8645 and 0.949895, respectively. The latter spends three times as many loop measurements and is not an algorithm-only gain.

V. Quantum Implementation and Scaling

For the recovered structured witness, inference is a genuine quantum measurement: measure the three-qubit Wilson parity directly, or accumulate it on an ancilla with three controlled-NOT gates and measure the ancilla. The projector $P_+ = (I + W)/2$ is then implemented without reconstructing a global density matrix. This is circuit compilation of a known Pauli string, not a quantum speedup.

For a general learned dense effect, diagonalization and basis-change synthesis are exponential in qubit count. A scalable quantum version therefore requires additional structure: a sparse Pauli/ribbon dictionary, stabilizer or tensor-network representation, a variational bounded observable, or a fault-tolerant block encoding and polynomial approximation to the sign function. The last route resembles quantum singular-value transformation but would require efficient coherent access to class states and fault-tolerant resources; we make no complexity advantage claim from the present simulator.

The most practical next test is not ideal-sector classification. It is surface-code syndrome drift [19]: update additive correlation states one round at a time, learn a low-rank gauge-compatible change witness, and ask whether it improves detection delay or decoder performance under equal total syndrome rounds. Marginal-rate CUSUM, covariance detectors, decoder likelihood, known stabilizer/Wilson diagnostics, and current correlation-aware drift methods must be included. Universal shadow-based sequential detection [20] is relevant when the measurement basis is not fixed; syndrome bits are already specific stabilizer readouts and must not be mislabeled as classical shadows.

VI. What the Results Do and Do Not Establish

The exact Run 4 result establishes:

- an observable-access advantage of a noncontractible algebra over every fixed sub-distance regional algebra;
- a symmetry-prior advantage over untwirled one-representative learning;
- recovery of a Wilson-equivalent code-space effect and an exact certificate when the declared local access contains zero label information.

It does not establish superiority over Helstrom discrimination, a Wilson-loop oracle, or logistic/threshold inference given the same parity feature. It does not establish quantum acceleration, universal sample efficiency, scalable tomography, a new Wilson loop or toric-code theorem, confinement, string theory, or holographic duality. The local-indistinguishability mechanism is known topological-order/QEC structure [6], [7]; the contribution is a reproducible integration of no-go certification, witness recovery, correct/wrong symmetry controls, and noise calibration.

Likewise, the broader repository results support matched-structure interpretation rather than universal dominance: Fourier-power logistic ties or slightly exceeds the invariant method, optical discrimination reaches the known Helstrom bound, and chemistry/robotics remain controlled models. A practical advantage must be established against the strongest non-oracle baseline using identical training, calibration, shot, and test budgets.

VII. Conclusion

The central object is not an unconstrained feature but an effect in a declared observable algebra. Conditional expectation projects class difference into that algebra, its positive part gives the optimal bounded witness, sector blocks explain where the difference resides, and additive class states admit streaming and distributed updates. Run 3 verifies this logic across four controlled domains. Run 4 adds a sharp topological test: the formalism first returns an exact local no-information certificate and then recovers the minimal Wilson-flux support when the access model permits it. The result is mathematically exact and operationally implementable, while its advantage remains one of representation and physical access. Circuit-level surface-code drift under equal budgets is the next falsifiable application.

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